



iSEEDx: Enhanced Intelligent Student Engagement and Affective State Analysis

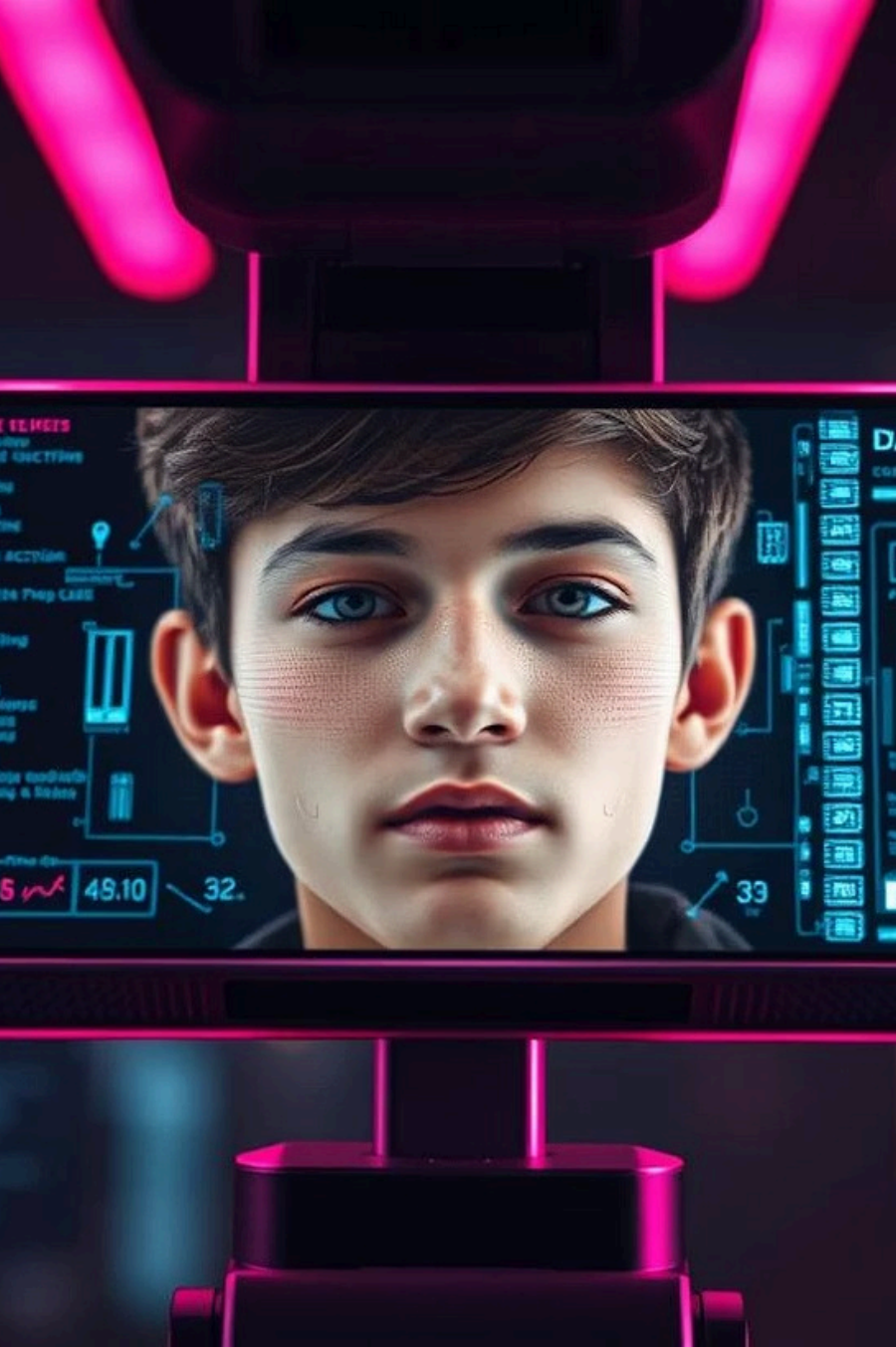
Project Title and Definition

**iSEEDx – Intelligent Student Engagement
Evaluation and Distraction eXtraction**

iSEEDx as a real-time AI-driven system that evaluates student focus and behavior using facial cues via webcam and adaptive self-learning sy

Objectives

- 1 Detect and classify focus levels using facial features and emotions
- 2 Track engagement percentage over time
- 3 Generate adaptive MCQs from uploaded study material
- 4 Send alerts when student is distracted
- 5 Visualize performance via a dashboard

A young man with dark hair and blue eyes is looking directly at the camera. He is positioned in front of a screen displaying a futuristic, high-tech interface with various data points, graphs, and icons. Above his head, a robotic head with glowing red eyes is visible. The overall scene is set in a dark environment with red and blue lighting.

Scope of the Project

- Detect multiple affective states (drowsiness, yawning, nodding, engagement) in real time.
- Personalize attentiveness monitoring based on user-uploaded study content.
- Provide real-time distraction alerts and motivational prompts during study sessions.
- Generate content-specific quizzes to assess post-study comprehension.
- Deliver detailed post-session analytics linking attention and quiz performance.
- Ensure ethical data collection with informed consent and anonymization.



Seamless User Journey



Start Session

User enters name, session purpose, and uploads documents for personalized monitoring setup.



Real-Time Monitoring

Drowsiness detection tracks user alertness during study, work, or reading sessions with live analytics.



Quiz & Analysis

Auto-redirect to PhD-level quiz generation and intelligent concept clustering for knowledge reinforcement.



Session Completion

Automatic file cleanup, session summary, and secure data disposal with comprehensive analytics report.

The intelligent clustering system discovers hidden relationships in session documents, building knowledge maps with causal links and dependencies across topics.

Literature Gaps Addressed by Our Enhanced Model



Absence of Content-Driven Personalization

Prior models miss tailoring analytics and feedback to user-supplied study material, focusing on generic engagement instead of specific content focus.



Lack of Just-in-Time Formative Assessment

Existing systems don't measure learning outcomes with dynamically generated, content-specific quizzes during study sessions, making feedback less actionable.



No Closed-Loop Feedback on Learning Outcomes

Systems rarely connect monitoring, active learning, and demonstrated understanding, missing insights into how focus behaviors correlate with assessment success.



Omission of Task-Goal Context in Analytics

Current models fail to account for user motivation, material complexity, or learning goals, limiting reporting on productive session parts or content mastery.

Advanced Drowsiness Detection System

Enhanced Eye Detection

EAR threshold optimized to 0.30 for earlier detection. Response time improved by 0.5 seconds with 8 consecutive frames.

Subtle Head Movement

Nodding threshold reduced to 15 pixels to catch subtle head droops. Triggers after just 5 movements.

Gradual Scoring System

Multi-level scoring replaces binary detection. Score accumulates based on drowsiness severity and triggers at score > 15 .

The system uses rolling averages across 10 frames to reduce false alarms while maintaining natural detection patterns. Active resets prevent score buildup when users are clearly alert.



Technologies and Models Used

Function	Dataset(s) Used	Use in Project
Eye open/closed	CEW, ZJU, mEBAL2	Train classifier for eye states, blink detection
Facial landmarks, pose, gaze, mouth	AFLW, 300W(-LP)	Landmark detection for gaze/yawn/head pose
Sequence/temporal blink detection	ZJU, mEBAL2	Time series for blink/distraction analysis
Multi-affective state recognition for e-learning: detection and classification of boredom, engagement, confusion, and frustration in real online learning scenarios	DAiSEE	Train and evaluate models that classify student affective states
Currently using dataset	yawn-dataset	

Interactive Document Features and Parameters



Smart Preview System

- Real-time preview of PDF pages, DOC content, and PPT slides
- Multi-format text and visual support with expand/collapse sections
- Integrated session timer with countdown and break reminders
- Visual progress tracking with completion percentages



Document Interaction Tools

- Highlighter for marking key text sections directly in preview
- Zoom controls with fit-to-screen and scaling options
- Sticky notes linked to specific pages or document sections
- Secure download with comprehensive session logging



EAR Analysis

- Eye Aspect Ratio threshold of 0.27 with consecutive frame monitoring across 10 frames for accurate detection.



Head Tracking

- Nodding detection with 15px threshold, triggering alerts after 5+ movements for comprehensive monitoring.



Smart Scoring

- Dynamic scoring system: $EAR < 0.27$ (+3 points), triggers at score > 20 with $EAR < 0.32$ condition.

Focus Percentage Computation in Our Enhanced Model



Behavioral Focus Percentage

Classifies webcam frames as "Focused" or "Distracted" using eye aspect ratio, head pose, gaze, and facial landmarks.

Formula:

$$\text{FocusPercentage} = \frac{\text{Focused seconds}}{\text{Total monitored seconds}} \times 100$$



Quiz/Preparedness Score

Administers a content-specific quiz post-session to objectively assess comprehension of the material.

Formula:

$$\text{PreparednessScore} = \frac{\text{Correct quiz answers}}{\text{Total questions}} \times 100$$



Combined Analytics

Displays both Focus Percentage and Quiz/Preparedness Score in a comprehensive session report.

Advanced analytics link attention drops to lower quiz performance on specific content, providing actionable insights.



Attentiveness Index (Composite Metric)

This composite metric integrates multiple emotional states to provide a more nuanced and richer assessment of a student's attentiveness beyond a simple focused/distracted classification.

Formula:

$$Ai = -0.598 \times \text{Boredom} + 1.539 \times \text{Engagement} - 0.085 \times \text{Yawning}$$

How Our Model Is Better



Direct Content & Goal Integration

Your system personalizes monitoring and analytics to user-uploaded content, reflecting real-world learning contexts and enabling richer datasets linking specific content areas with attention metrics.



Content-Aware Quizzes

Immediately after a study session, the model generates a targeted quiz from self-uploaded content, providing a direct, objective measurement of user comprehension and retention, overcoming generic quiz approaches.



Actionable Closed-Loop Analytics

By combining real-time behavioral monitoring (focus, distractions) with quiz performance, users gain clear insight into how their focus habits impacted learning outcomes, such as attentive periods correlating with better answers.



Adaptive Goal-Linked Alerts

During a session, "Stay focused!" prompts are tuned to the complexity or criticality of the content being studied, rather than relying on generic timing or distraction signals, making them more relevant and effective.



Comprehensive Post-Session Review

After a session, users receive contextualized analytics showing challenging content areas, attention shifts relative to task segments, and personalized tips for future learning, moving beyond simple summary stats.



New Research Paradigms Enabled

Our model yields a unique dataset connecting content, attention patterns, and verified outcomes, facilitating deeper research into which behaviors genuinely drive successful learning in diverse contexts.

System Achievements & Future Vision

Current Benefits

- Natural flow from monitoring to quiz insights
- Enhanced engagement keeps users active and aware
- Accurate detection prevents false triggers
- Learning value through post-session reinforcement

Export Capabilities

Results available in JSON, CSV, and PDF formats for comprehensive analysis and record-keeping.

Next Release

Multi-language quizzes,
video analysis, mobile
apps



Final Outcome

1

Real-time Engagement Monitoring

System for continuous student engagement tracking.

2

Live Focus % Computation

Instantaneous calculation and display of focus levels.

3

Study Efficiency Measurement

Tools to quantify and improve learning effectiveness.

4

Teacher/Student Dashboard

Comprehensive dashboard with trends and insights.